THERAGRIP

AI-Assisted Instrumented Grip-Training and Rehabilitation Measurement Device

Theme: Health Tech — Innovating Healthcare, Assistive Devices & Rural Care

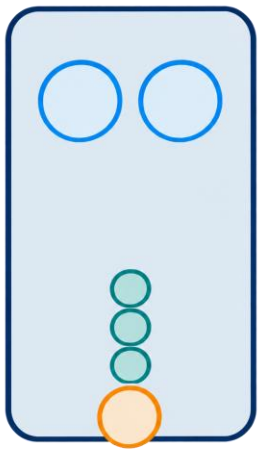
1. TITLE OF THE PROJECT / PRODUCT

TheraGrip — AI-Assisted Instrumented Grip-Training and Rehabilitation Measurement Device

Tagline:

Measure. Move. Improve.

TheraGrip is a compact physical grip-training prototype that combines localized force sensing, motion sensing and AI-assisted analysis to provide objective, repetition-level feedback during structured hand and grip exercises.



2. PROBLEM STATEMENT

Problem Statement

Conventional hand and grip rehabilitation exercises often rely on repetitive practice and intermittent observation, making it difficult to continuously measure grip interaction, movement consistency and repetition quality and to provide immediate, objective feedback during every exercise.

Why this problem matters

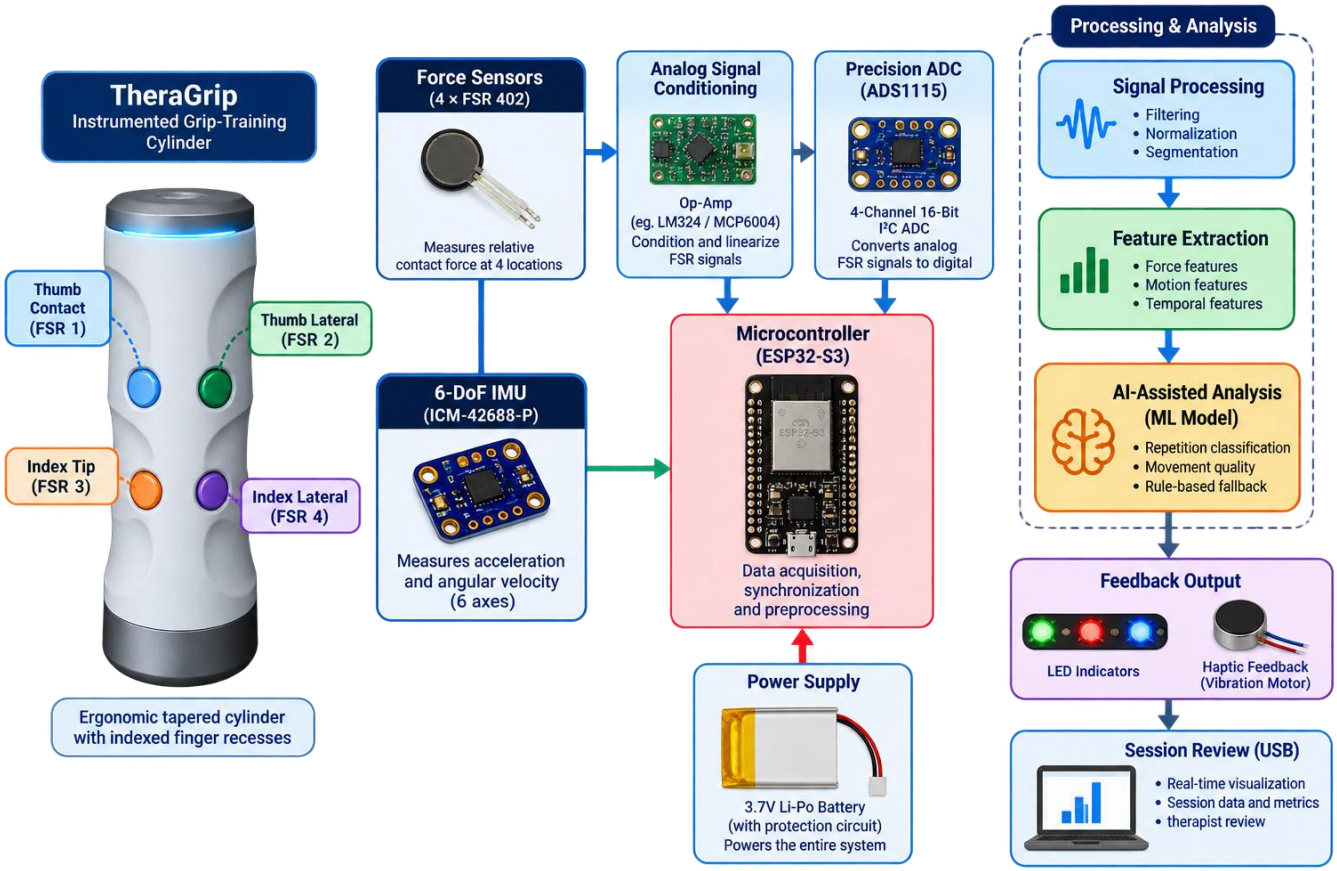
Hand rehabilitation can require repeated exercises over many sessions. In conventional practice, a person may use a stress ball, therapy putty, grip trainer or similar basic exercise object, while the therapist observes only selected repetitions during a session.

This creates several practical limitations:

- Force application may vary between repetitions.
- Finger contact and grip distribution are difficult to observe continuously.
- Movement deviations may occur without immediate feedback.
- A therapist cannot continuously observe every repetition.
- Progress may be recorded through periodic assessments rather than objective repetition-level measurements.
- Basic exercise tools generally do not capture synchronized information about **how the user grips and how the device moves during the exercise**.

TheraGrip addresses this measurement gap by making the exercise object itself an instrumented hardware system.

3. SOLUTION PROPOSED



3.1 Proposed Solution

TheraGrip is an **ergonomically constrained, tapered cylindrical grip-training device** designed to create repeatable thumb and index-finger contact.

Unlike a generic soft ball, the proposed device uses defined contact regions so that the sensing geometry remains more consistent between repetitions.

TheraGrip's innovation lies in the integration of repeatable finger-contact geometry, localized multi-point force sensing, 6-DoF motion sensing and AI-assisted repetition

analysis into a compact instrumented grip-training device, enabling immediate feedback on exercise performance rather than relying solely on occasional subjective observation.

The primary prototype will use:

- A tapered cylindrical 3D-printed enclosure
- Defined thumb and index-finger contact recesses
- Four localized force-sensing regions
- A 6-DoF inertial measurement unit (IMU)
- ESP32-S3 microcontroller
- ADS1115 external ADC
- Analog signal-conditioning circuitry
- Rechargeable protected Li-Po battery
- Optional LED and vibration-based feedback
- A computer-based analysis dashboard during the initial development stage

The detailed engineering design specifies a roughly **130 mm cylindrical form factor**, with tapered grip geometry and mechanical registration areas for repeatable thumb/index placement.

3.2 How TheraGrip Works

The basic operating sequence is:

Grip → Sense → Process → Analyse → Feedback → Review

Step 1 — Grip

The user holds the TheraGrip device using the predefined thumb and index contact regions.

Step 2 — Force sensing

Four localized force-sensitive resistors (FSRs) capture changes in contact/loading patterns at different sensing zones.

The FSRs are treated primarily as **relative and multi-point sensing elements**, rather than as clinical-grade absolute force meters.

Step 3 — Motion sensing

A 6-DoF IMU captures device-level motion information such as:

- acceleration,
- angular velocity,
- orientation-related features,
- movement smoothness,
- jerk-related features,
- repetition consistency.

Step 4 — Signal processing

The ESP32-S3 collects the sensor data and performs preprocessing.

The system can apply:

- filtering,
- sensor normalization,
- feature extraction,
- temporal segmentation,
- movement-window analysis.

Step 5 — AI-assisted analysis

The system investigates whether combining multiple force and motion features can distinguish predefined exercise-performance patterns more effectively than simple threshold/rule-based methods.

Candidate models include lightweight tree-based machine-learning approaches such as:

- Random Forest
- Gradient-boosted decision trees

The AI component is therefore **experimentally validated rather than assumed to be automatically superior**.

Step 6 — Feedback

The device can provide immediate feedback through:

- LED indication,
- vibration/haptic feedback,
- Or a connected dashboard.

Feedback is intended to indicate measurable deviations such as:

- force-pattern deviation,
- movement-consistency deviation,
- high device tilt during a predefined movement,
- repetition inconsistency.

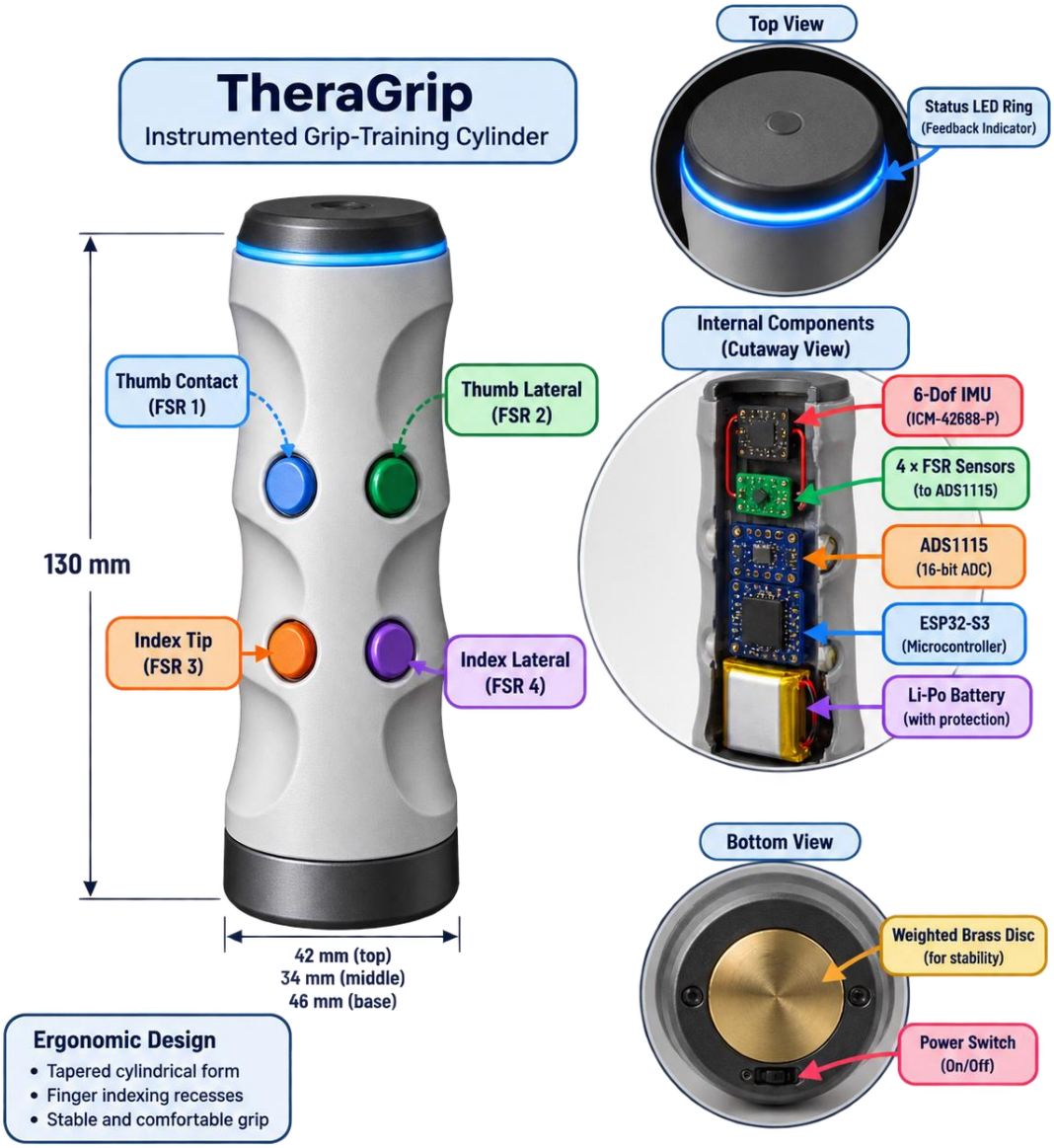
Step 7 — Therapist review

A dashboard can display session-level information such as:

- repetition count,
- force-pattern features,
- movement consistency,
- repetition-to-repetition variation,
- measured performance trends.

The system is intended to support observation and feedback, **not replace professional clinical judgment**.

3.3 Physical and Electronic Architecture



Hardware

The refined prototype architecture consists of:

User contact → silicone mechanical interface → FSR sensors → analog signal conditioning → ADS1115 ADC → ESP32-S3

alongside:

IMU → ESP32-S3 → signal processing → feature extraction → AI-assisted classification → feedback

The system uses a protected rechargeable battery enclosed within the device and isolated from the user's contact surfaces.

The engineering investigation specifically recommends an ESP32-S3, ADS1115, four FSR sensing zones, a 6-DoF IMU, haptic actuator and protected Li-Po power architecture.

3.4 Role of AI

AI is not included merely to display sensor readings.

The proposed validation compares:

Simple rule/threshold baseline

against

AI-assisted multimodal classification

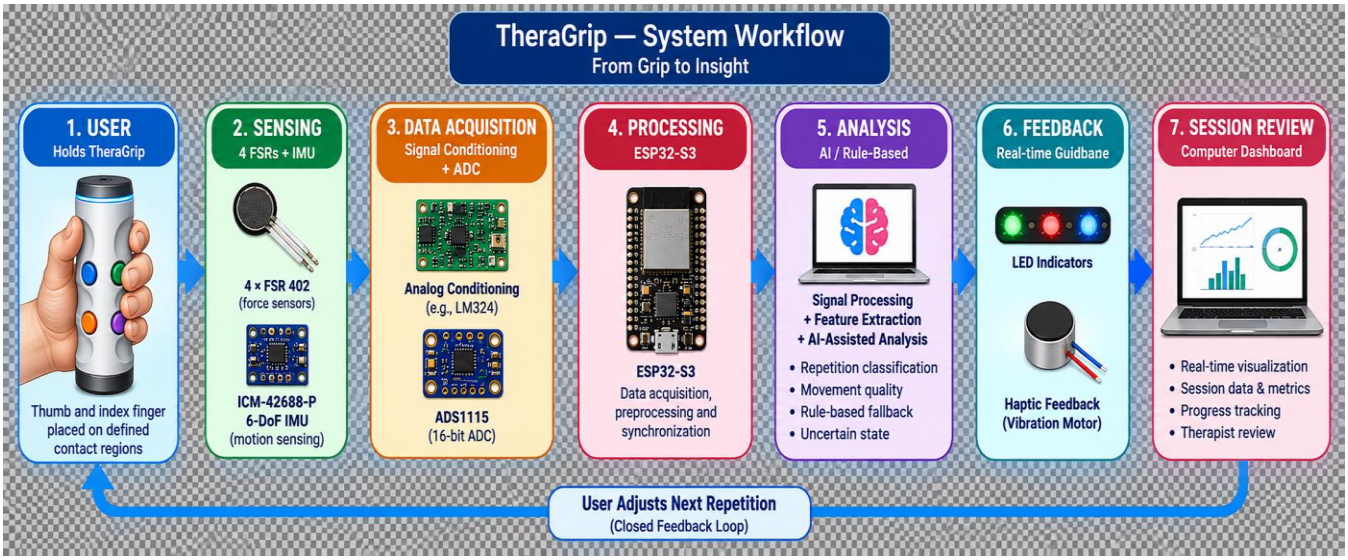
using combinations of:

- force distribution features,
- force timing,
- force rise/fall characteristics,
- motion features,
- angular velocity,
- acceleration,
- movement consistency,
- temporal relationships between force and motion.

The experiment will use **subject-independent validation**, meaning participants used for testing will be separated from those used to train the model.

Healthy volunteers will be used for controlled movement-pattern experiments and **will not be treated as substitutes for clinical patient populations**.

The system will also allow an **“uncertain”** output when the available sensor information is insufficient for reliable classification.



4. IMPLEMENTATION PLAN

4.1 Target Beneficiaries

The initial intended beneficiaries are:

Primary users

- Individuals performing structured hand and grip exercises
- People requiring repetitive grip-training practice
- Rehabilitation users who need objective exercise feedback

Professional users

- Physiotherapists
- Occupational therapists
- Rehabilitation professionals

Longer-term beneficiaries

The concept could eventually support rehabilitation centres, community healthcare environments, and home-based exercise programs, subject to further validation.

4.2 Development and Deployment Plan

Phase 1 — Engineering Prototype

Develop the primary indexed cylindrical TheraGrip device.

Tasks:

- CAD design
- 3D printing
- sensor integration
- analog electronics
- ESP32-S3 firmware
- IMU integration
- protected battery integration

Phase 2 — Sensor Validation

Evaluate:

- FSR repeatability
- sensor drift
- contact consistency
- ADC noise
- IMU signal quality
- timing between sensing channels
- mechanical stability

A reference load-cell/dynamometer measurement will be used where appropriate to validate the sensing behaviour.

Phase 3 — Data Collection

Collect controlled exercise data from volunteers.

The initial MVP will focus on **one or two predefined exercises**, rather than attempting to support a large rehabilitation library.

Participants will perform predefined movement patterns under controlled conditions.

Phase 4 — AI and Rule-Based Comparison

Compare:

1. simple rule-based detection,
2. force-only features,
3. force + motion features,
4. AI-assisted classification.

The purpose is to determine whether multimodal sensing provides measurable information beyond simple grip magnitude.

Phase 5 — Feedback Prototype

Integrate:

- LED feedback,
- vibration/haptic feedback,
- real-time dashboard,
- session logging.

Phase 6 — Validation and Demonstration

Evaluate:

- sensor repeatability,
- classification performance,
- subject-independent generalization,
- feedback response,
- usability,
- hardware reliability.

4.3 Expected Outcomes

The expected outcome is a functional physical prototype capable of:

- sensing localized grip interaction,
- capturing device motion,
- recording exercise repetitions,
- extracting measurable movement/force features,
- comparing predefined movement patterns,
- providing immediate feedback,
- generating session-level performance information.

The prototype will establish whether the proposed combination of **multi-point force sensing + motion sensing + AI-assisted analysis** provides useful information for structured grip-training exercises.

4.4 Expected Deliverables

Deliverable 1

Working TheraGrip physical prototype.

Deliverable 2

Embedded sensor acquisition and preprocessing firmware.

Deliverable 3

Validated force and motion data pipeline.

Deliverable 4

AI-versus-rule validation experiment.

Deliverable 5

Real-time feedback mechanism.

Deliverable 6

Session-level monitoring dashboard.

Deliverable 7

Validation report containing sensor, AI, and usability results.

5. SAFETY AND EFFICIENCY ANALYSIS

5.1 Safety

TheraGrip is designed as a **rehabilitation-assistance and measurement prototype**, not as a diagnostic or treatment device.

Mechanical safety

The enclosure will:

- isolate electronics from the user's hand,
- protect sensor wiring,
- avoid exposed conductive components,
- provide stable grip geometry,
- use rounded ergonomic surfaces.

The primary device will be mechanically constrained to prevent unnecessary movement of the sensing regions.

Electrical safety

The prototype will use:

- a protected rechargeable Li-Po battery,
- dedicated charging/protection circuitry,
- regulated low-voltage electronics,
- an enclosed battery compartment,
- insulated wiring,
- physical power switching.

Sensor safety

FSRs will not be directly exposed as bare sensing films against the user's skin.

A compliant mechanical interface will distribute contact forces and protect the sensor.

AI safety

AI outputs will be treated as **advisory engineering feedback**, not medical diagnoses.

The system will:

- avoid diagnostic labels,
- include an uncertain state,
- maintain a rule-based fallback,
- log sensor-health conditions,
- avoid presenting experimental thresholds as clinical limits.

Data privacy

The prototype will minimize personally identifiable information.

User records can use pseudonymised identifiers rather than names, and raw sensor data will be stored separately from identifying information.

5.2 Efficiency

TheraGrip improves the efficiency of the rehabilitation measurement process by reducing dependence on continuous manual observation.

Instead of relying only on:

Exercise → occasional observation → subjective assessment

the proposed workflow becomes:

Exercise → sensing → automated measurement → immediate feedback → session data → review

This allows each repetition to contribute measurable information.

The system is therefore intended to improve **measurement efficiency and feedback consistency**, rather than claiming to replace a therapist.

6. SCALABILITY AND FUTURE DEVELOPMENT

The initial prototype intentionally focuses on **one primary indexed cylindrical device**.

This reduces mechanical and sensing variability during early validation.

Once the core sensing architecture has been validated, the platform can be expanded.

Future Development 1 — Additional Grip Geometries

Future versions may include:

- pinch-oriented attachments,
- different grip diameters,
- task-specific rehabilitation interfaces.

However, each attachment would require its own mechanical calibration and validation.

Future Development 2 — Improved Sensing

Future versions may investigate:

- higher-accuracy force sensors,
- calibrated load cells,
- improved mechanical coupling,
- additional sensing modalities.

Future Development 3 — Edge AI

After validation on a computer-based pipeline, lightweight models can be optimized for embedded inference on the ESP32-S3.

Edge inference could reduce dependence on an external computer.

Future Development 4 — Wireless Connectivity

Wireless connectivity can be added after the USB-first prototype is stable.

This could support:

- remote session transfer,
- therapist dashboards,
- longitudinal data storage.

Future Development 5 — Rehabilitation Exercise Library

The system could eventually support multiple predefined exercises with separate validated models.

Each exercise would have its own:

- movement definition,
- sensing configuration,
- calibration procedure,
- performance metrics.

Future Development 6 — Clinical Validation

A future stage can involve collaboration with rehabilitation professionals and institutions to evaluate the system with appropriate ethical oversight and clinical protocols.

Clinical validation is **not claimed at Stage 1**.

7. INNOVATION ADVANTAGE

7.1 What is innovative about TheraGrip?

The innovation is not simply the use of an ESP32, sensors or an AI dashboard.

The proposed innovation lies in combining:

Repeatable physical interaction

with

localized multi-point force sensing

and

device-level motion sensing

and

AI-assisted analysis of their temporal relationship

within a compact physical rehabilitation-training object.

The device therefore attempts to measure **how the user interacts with the grip and how that interaction changes during a movement**, rather than recording only a single grip-strength value.

7.2 Why Four Localized Sensors?

A key engineering question is:

Why use multiple localized sensing regions instead of a single force sensor?

The proposed answer is that a single force measurement primarily represents overall loading, while multiple sensing regions can provide information about **where and when contact/loading occurs**.

TheraGrip will therefore experimentally compare:

- one-force-sensor representation,
- two-sensor representation,
- four-localized-sensor representation,

to determine whether additional spatial information improves discrimination of predefined exercise patterns.

This is a proposed validation experiment, not an unsupported claim that four sensors are automatically superior.

7.3 Differentiation from Existing Solutions

Existing rehabilitation and grip-monitoring products demonstrate that instrumented grip sensing, motion sensing and therapist-facing software are already established areas.

TheraGrip therefore does **not** claim to be the first device to combine force and motion sensing.

The competitive landscape research identified existing commercial and research systems, including GripAble, Squegg, Jamar Smart Digital and several robotic/wearable rehabilitation platforms. It also identified relevant Indian research/prior-art involving force and motion sensing.

TheraGrip's proposed differentiation is instead based on the combination of:

- constrained and repeatable finger-contact geometry,
- localized multi-point force interaction,
- motion dynamics,
- explicit AI-versus-rule validation,
- low-cost prototype-oriented hardware,
- immediate physical feedback,
- and a focus on measurable repetition-level exercise behaviour.

The project also explicitly recognizes the overlap with existing technologies and prior art rather than claiming “first” or “no competition.”

7.4 Innovation Validation

The innovation will ultimately be evaluated experimentally through:

Experiment A

Sensor repeatability.

Experiment B

One-sensor versus multi-sensor information.

Experiment C

Force-only versus force + motion features.

Experiment D

Rule-based versus AI-assisted classification.

Experiment E

Subject-independent validation.

This makes the innovation claim testable rather than purely descriptive.

8. VISUAL IMPACT & USER JOURNEY



8.1 Before → After

BEFORE — Conventional Exercise

1. User performs repetitive grip exercises.
2. Basic exercise object provides little measurement.
3. Therapist cannot continuously observe every repetition.
4. Force and movement quality are difficult to quantify.
5. Deviations may go unnoticed.
6. Progress is reviewed periodically.

AFTER — With TheraGrip

1. User picks up TheraGrip.
2. Defined contact regions establish repeatable interaction.

3. Force sensors and IMU capture data.
4. ESP32 processes the sensor information.
5. The system analyses repetition-level force and motion features.
6. The device provides immediate visual/haptic feedback.
7. Session measurements are available for therapist review.
8. Performance metrics can be compared across repeated sessions.

8.2 User Journey

User → Grip → Sense → Analyse → Feedback → Adjust → Review → Track

User starts exercise

↓

TheraGrip establishes repeatable grip interaction

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FSRs capture localized force patterns

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IMU captures device motion

↓

ESP32 processes sensor data

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AI/rule-based analysis evaluates predefined movement patterns

↓

LED/haptic feedback is provided

↓

User adjusts the next repetition if needed

↓

Session measurements are stored

↓

Therapist reviews measured performance

↓

Performance changes can be compared over time

The storyboard in your existing visual set already communicates the core sequence from exercise → sensing → ESP32 → AI evaluation → feedback → therapist review.

Conclusion

TheraGrip proposes a physically instrumented approach to grip-training measurement in which the exercise device itself becomes a sensing and feedback interface.

By combining repeatable finger-contact geometry, localized multi-point force sensing, motion sensing and AI-assisted analysis, the project aims to investigate whether repetition-level exercise behaviour can be measured more objectively than with conventional passive grip-training tools.

The Stage 1 prototype will focus on a single indexed cylindrical device, controlled exercises, sensor validation and AI-versus-rule comparison. Future development can expand the platform after the core sensing and classification approach has been experimentally validated.

TheraGrip is therefore positioned as a rehabilitation-assistance and measurement prototype—not a diagnostic or treatment device.